Berea Technical College

Department of Engineering Studies

Digital Electronics N4
Assignment 1

QUESTION 1: NUMBER SYSTEMS

1.1 Subtract each of the following by means of the one's complement and convert

The final answer to the decimal system:

1.1.1 $-58,75_{10} - 46,5_{10}$

1.1.2 $C, A_{16} - 7,B_{16}$

1.2 Calculate the following in their binary form and convert the final answer to the

Decimal number system:

1.2.1 $6A,5_{16} \div 3,7_{8}$

1.2.2 $6D, A_{16} \times 55,5_{8}$

1.3 Convert the following codes as indicated:

1.3.1 101101_{2} to Gray

1.3.2 $28,4_{10}$to XS3

1.3.3 $2A,F8_{16}$ to Binary

QUESTION 2: FURTHER CODES AND LOGIC GATES

2.1 What is meaning of BCD?

2.2 Name any non-Inverting code

2.3 Give the symbol, the truth table and Boolean expression of an AND gate.